Advanced Causal Inference

Lecture 01: Introduction

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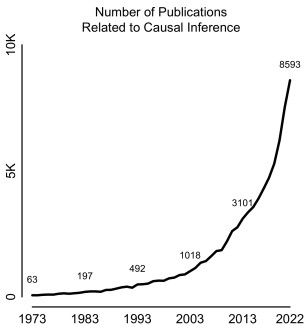
This lecture note is partially based on the lecture notes/textbook of Ying Zhou

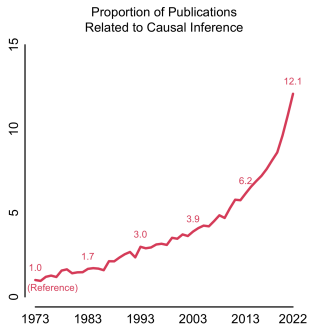
Causal Inference

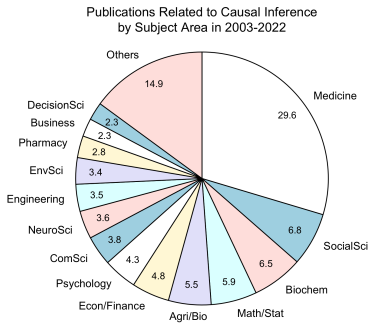
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Causal Inference

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- Why/How can causal inference be useful?

Causal Inference

- We have **big data** and **machine learning** techniques
- Why/How can causal inference be useful?
- Most machine learning algorithms depend on correlations
  - NLP
  - Self-driving car
  - ImageNet
  - ....
- Correlations alone are a dangerous path to actionable insights

How do prediction systems work?

- **Aim:** Suppose a doctor wants to predict a patient's probability of survival.
- Collect data on many attributes:
  - chronic diseases, the duration of bed rest, diet, etc.
- Build a predictive model

$$\text{Probability of survival} = f(\text{chronic diseases, the duration of bed rest, diet})$$

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- Or better yet, don't sleep at all — just sleep on the floor?!

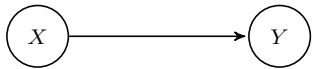
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Confounding!

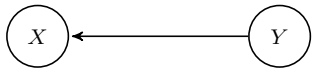
Sources of Correlation

- Causation



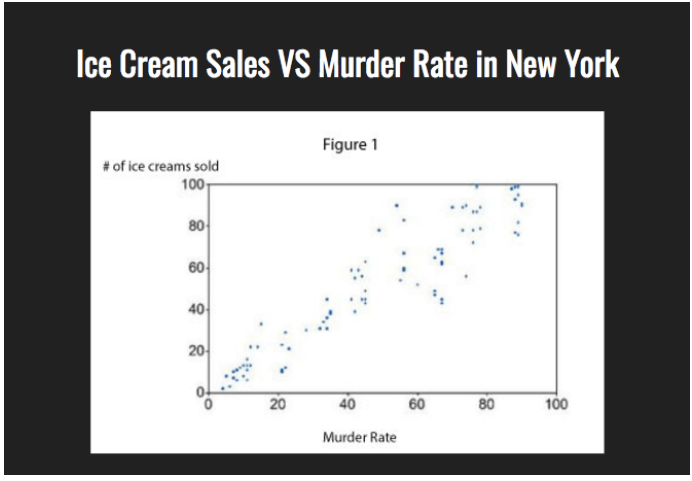
Sources of Correlation

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- Reverse causation



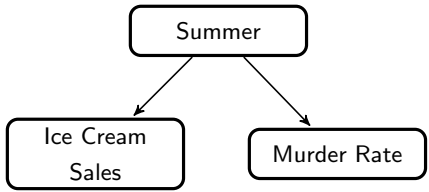
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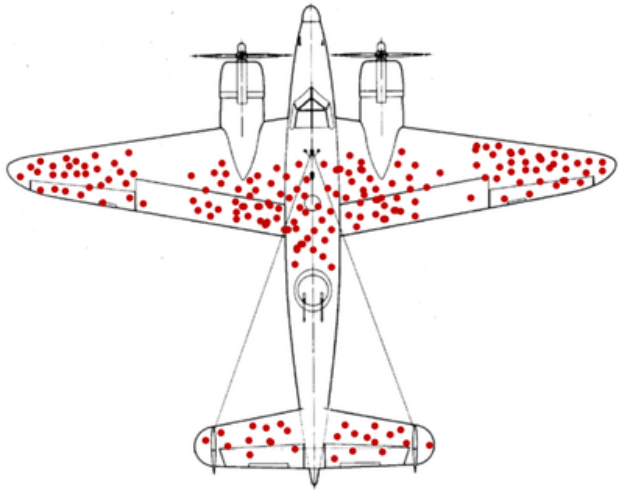
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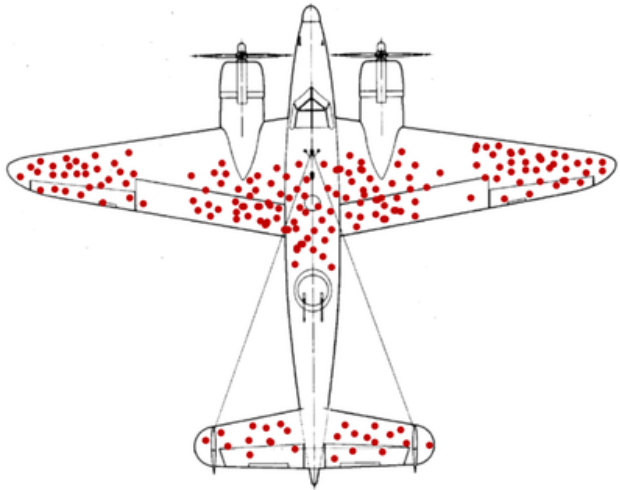
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Sources of Correlation

- Causation
- Reverse causation
- Confounding
- Selection bias (Wald, 1943)
- Only causation leads to actionable insights!



correlation lets you predict the future

causality lets you change the future

Disclaimer of This Course

- Understanding causality roughly consists of

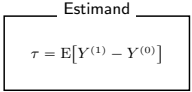
  1. **Causal discovery**
     - process of uncovering cause-and-effect relationships between variables in a dataset, often represented as a directed acyclic graph (DAG)

  2. **Causal identification**
     - process of assessing whether the causal effect of one variable on another can be uniquely identified from observed data and the specified causal model (e.g., DAG, structural equation models)

  3. **Causal inference**
     - process of statistically inferring the causal effect of one variable on another from observed data and the specified causal model

- The class will primarily focus on 3, cover some topics in 1 and 2

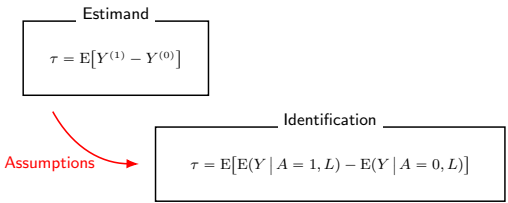
Flow of Causal Inference



Define an estimand

(e.g., Average Treatment Effect (ATE))

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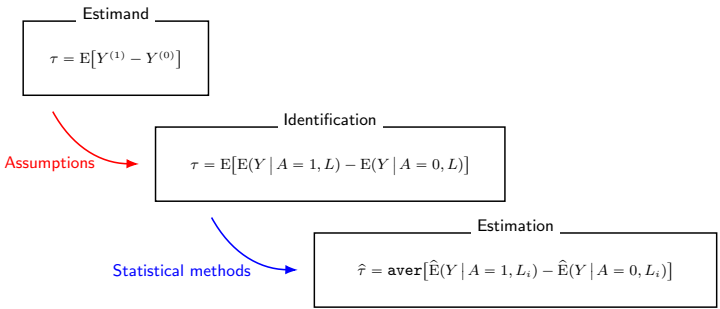
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Establish identification

(e.g., g-formula, IPW)

Flow of Causal Inference



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(e.g., Average Treatment Effect (ATE))

Establish identification

(e.g., g-formula, IPW)

Construct an estimator

(e.g., OLS, Machine learning)

References

Wald, A. (1943). A method of estimating plane vulnerability based on damage of survivors. *Statistical Research Group, Columbia University. CRC*, 432:159.